IBRAHIM TIJANI, PhD, P.Eng., PMP

Mechanics of Materials & Infrastructure | Composites | Data-Driven Mechanics | SHM

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RESEARCH PROFILE

Civil/structural engineering scholar and Canadian Professional Engineer with research spanning composite mechanics, FRP-confined and strengthened concrete, interface/fracture behavior, advanced construction materials, structural health monitoring, non-destructive evaluation, and data-driven mechanics. Postdoctoral research experience at UBC and The Hong Kong Polytechnic University, with a record of peer-reviewed publications, externally funded research, interdisciplinary collaboration, and translation of ML/sensing methods into engineering decision-support tools. Current research direction integrates mechanics, distributed sensing, reduced-order/data-driven modeling, explainable AI, and digital twins for resilient infrastructure and energy-relevant composite systems.

CORE EXPERTISE

- Composite and structural mechanics; FRP confinement/strengthening; interface fracture and bond
- Nonlinear structural behavior; damage/degradation; experimental and numerical mechanics
- Structural health monitoring; NDE; digital image correlation; acoustic/vibro-acoustic sensing
- Machine learning; explainable AI; surrogate modeling; CTGAN data augmentation; optimization
- Research coordination; industry collaboration; technical reporting; project delivery (PMP)
- Professional engineering practice and infrastructure assessment (P.Eng., Manitoba)

ACADEMIC APPOINTMENTS

2026-present Research Coordinator, School of Engineering, University of British Columbia (UBC), Canada
Coordinate multidisciplinary research planning, experimental/data activities, milestones, proposals, manuscripts, technical reports, and collaboration. Contribute to data analysis, funding applications, and research dissemination.

2022-2026 Mitacs Accelerate Industrial Postdoctoral Fellow, School of Engineering, UBC, Canada
Research on intelligent and non-destructive solutions for structural/urban infrastructure; ML frameworks, engineering tools, and industry-linked research outputs.

2020-2022 Postdoctoral Fellow / Project Coordinator, The Hong Kong Polytechnic University, Hong Kong SAR
Smart infrastructure monitoring, leak diagnosis, sensing/data-driven engineering, project coordination, reporting, and grant-related documentation.

2017-2020 Research & Teaching Assistant / PhD Candidate, City University of Hong Kong
Experimental and numerical studies of FRP-repaired and FRP-confined concrete; undergraduate teaching and laboratory support.

2014-2017 Teaching Assistant / MSc Student, King Saud University, Saudi Arabia
Teaching and assessment support in construction engineering and management, planning/control, and cost analysis.

PROFESSIONAL EXPERIENCE

2023 Project Engineer / Assessment Officer, Government of Manitoba, Canada

Engineering/infrastructure assessment, site reviews, technical analysis, drawings/records review, reporting, and mentoring.

2022-2023 Project Engineer in Training, Spannovation Consulting Ltd., British Columbia, Canada

Structural design reviews and design-check spreadsheets for steel flexure, shear, compression, and tension.

EDUCATION

2024 Certificate, Water Resources Management and Policy, Université de Genève
2020 PhD, Civil Engineering (Structural Engineering), City University of Hong Kong
2017 MSc, Civil Engineering, King Saud University, Saudi Arabia (GPA 4.50/5.00)
2011 BTech, Civil Engineering, Ladoke Akintola University of Technology, Nigeria

LICENSURE, CERTIFICATIONS & MAJOR FELLOWSHIPS

- Registered Professional Engineer (P.Eng.), Engineers Geoscientists Manitoba, Canada (2023)
- Project Management Professional (PMP), Project Management Institute (2023)
- Mitacs Accelerate Industrial Postdoctoral Fellowship, UBC (2022-2026)
- Innovation and Technology Fund Postdoctoral Fellowship, Hong Kong SAR (2020-2022)
- Hong Kong PhD Fellowship Scheme, Research Grants Council of Hong Kong SAR (2017-2020)
- Postgraduate Scholarship, Ministry of Education, King Saud University (2014-2017)

SELECTED RESEARCH PROJECTS

2022-present Precast system with high-strength lightweight green concrete - Mitacs Accelerate, Canada; supervisor: Prof. M. Shahria Alam.

2020-2022 Smart noise logger-based leak diagnosis system - Innovation and Technology Fund, Hong Kong SAR; supervisor: Prof. Tarek Zayed.

2017-2020 Mechanical relationship of concrete with uniform and non-uniform confinement - Research-grant-supported PhD work; supervisors: Prof. C.W. Lim, Prof. Yu-Fei Wu, Dr. Cheng Jiang.

TEACHING & SUPERVISION EXPERIENCE

Courses taught/supported across PolyU, CityU Hong Kong, and King Saud University include Project Management, Construction Administration, Engineering Surveying & Photogrammetry, Quantity Surveying, Construction Law, Applied Research Project, Structural Analysis, Steel Design, Fluid Mechanics, Hydraulics and Hydrology, Foundation Engineering, Geotechnical Engineering, Engineering Drawing, Highway Design, Computer Applications in Civil Engineering, Construction Management, Construction Planning and Control, and Cost Analysis and Control. Responsibilities included tutorials, lesson/course preparation, grading, student evaluation, laboratory support, final-year project supervision, and curriculum support.

PEER-REVIEWED JOURNAL PUBLICATIONS

24 peer-reviewed journal articles | 17 Q1 | 6 Q2 | 23/24 (96%) published in Q1 – Q2 journals

1. **Tijani I.A**, Haroglu H., Wakjira T.G, Abadel A., & Alam M.S (2026). Data-augmented explainable machine learning framework and user-centric tool for classifying failure modes and predicting flexural capacity of FRP-strengthened RC beams. *Structures*. 111776. [Q1].
2. **Tijani I.A**, Kanaan D, Wakjira T.G, Alam M.S, & Haroglu H (2026). Explainable machine learning in concrete technology: A state-of-the-art review with future innovations. *Archives of Computational Methods in Engineering*. 33, 6095-6123. [Q1].
3. **Tijani I.A** & C Jiang. (2025) GUI-based hybrid ML model for predicting ultimate strength of FRP-confined UHPC with CTGAN-augmented data. *Composite Structures*. 119905. [Q1].
4. **Tijani I.A**, T.G Wakjira, M.S Alam, & N Uddin. (2025) Digital image correlation for structural health monitoring of bridge systems: A state-of-the-art review with future research directions. *Archives of Computational Methods in Engineering*. 33, 4325-4341. [Q1].
5. Zhang R, Fares A., **Tijani I.A**, Zayed T. Jin Z, and Yussif A. (2025) Acoustic based leak location detection for water supply pipelines in urban areas via multi-task deep learning. *Engineering Applications of Artificial Intelligence*. 1112718. [Q1].
6. **Tijani I.A**, T.G Wakjira, H. Haroglu & M.S Alam. (2025). A surrogate-based machine learning model for prediction of interfacial fracture energy in externally bonded FRP–concrete bonds with a user-friendly tool. *Engineering Fracture Mechanics*. 111633. [Q1].
7. **Tijani I.A**, T.G Wakjira, H. Haroglu & M.S Alam. (2025) Explainable machine learning and application-oriented tool for predicting effective hoop strain of FRP-confined concrete. *Frontiers of Structural and Civil Engineering*. 19, 1621-1636. [Q1].
8. Wakjira, T.G., **Tijani, I.A**., Alam, M.S., Mashal, M., & Hasan, M.K. (2025) Can we trust AI content detection tools for critical decision-making? *Information*, 16, 904. [Q2].
9. Liu, R., Tariq, S., **Tijani, I.A**, Fan, H, Abdelmageed, S Fares, A., & Zayed, T. (2025). Leak detection in water distribution networks using micro-electromechanical systems-based accelerometers: a machine learning approach. *Water Practice and Technology*. 20(9), 1900-1920. [Q2].
10. **Tijani I.A**, J.A Dauda, M.A Kareem, & A.I Lawal. (2025). Genetic programming-based model for estimating maximum pull load of FRP-to-concrete bond interfaces with GUI implementation. *Structural Concrete*. 26(6), 8899-8912 [Q1].
11. Zhang, R., Yussif, A., **Tijani, I.A**., Fares, A., Tariq, S., & Zayed, T (2024). Acoustic localization approach for urban water distribution networks using machine learning method. *Engineering Applications of Artificial Intelligence*. 137. 109062. [Q1].
12. Liu, R., Tariq, S., **Tijani, I.A**., Fares, A., Bakhtawar, B., Fan, H., Zhang, R., & Zayed, T. (2024). Data-driven approaches for vibroacoustic localization of leaks in water distribution networks. *Environmental Processes.* 11, 14. [Q1].
13. **Tijani, I.A**., Lawal, A.I., Ogunsola, N.O, & Kwon, S. (2023) Prediction of ultimate strength of FRP-confined predamaged concrete using backward multiple regression motivated soft computing methods. *Scientia Iranica*. e23225. [Q3].
14. Kareem, M.A., Akintonde, B.B., Adesoye, J.S., Ishola, K., **Tijani, I.A.**, & others. (2023). Influence of Cashew Leaf Ash as Partial Replacement for Cement on the Properties of Fresh and Hardened Concrete. *Cleaner Waste Systems*. 4, 100063. [Q2].
15. **Tijani, I.A.**, Lawal, A.I., & Kwon, S. (2022). Machine learning techniques for prediction of ultimate strain of FRP-confined concrete. *Structural Engineering and Mechanics* 84(1):101-111 [Q2].
16. **Tijani I.A** & Zayed T. (2022). Gene Expression Programming Based Mathematical Models for Leak Detection of Water Distribution Networks. *Measurement*, 188:110611 [Q1].
17. Dauda, J.A., Rahmon, S., **Tijani, I.A.,** & Mohammed, F. (2022). Design Optimisation of Reinforced Concrete Pile Foundation using Generalized Reduced Gradient Algorithm. *Frontiers in Engineering and Built Environment* 2(3):133-153. [Q2].
18. **Tijani I.A**., Abdelmageed, S., Fares, A., Fan, K.H., Hu, Z.Y., & Zayed, T., (2022). Improving the leak detection efficiency in water distribution networks using noise loggers. *Science of the Total Environment*. 821, 153530. [Q1].
19. Fares, A., **Tijani, I. A** & others (2022) Leak detection in real water distribution networks based on acoustic emission and machine learning. *Environmental Technology*. 44(25), 3850-3866. [Q2].
20. **Tijani I.A**., Jiang, C., Lim. C.W., & Wu, Y.F. (2021). Eccentrically loaded concrete under nonuniform passive confinement. *ASCE Journal Structural Engineering*. 04021247. [Q1].
21. **Tijani I.A**., Wu, Y.F., & Lim, C.W. (2021). "Effects of pre-damage on stress-strain relationship of partially confined concrete." *ACI Structural Journal*. 118(1):61–72. [Q1].
22. **Tijani I.A**., Jiang, C., Lim. C.W., & Wu, Y.F. (2020). Effect of load eccentricity on the mechanical response of FRP-confined pre-damaged concrete columns. *ASCE Journal Composites for Construction*, 04020057. [Q1].

23. **Tijani I.A** & others (2020). "Energy balance method for modeling ultimate strain of fiber-reinforced polymer-repaired concrete." *Structural Concrete*, 21(2):804–820. [Q1].
24. **Tijani I. A** & others (2019). "Aggregate size effects and general static loading response on mechanical behavior of passively confined concrete." *Construction and Building Materials*, 205:61–72. [Q1].

PROFESSIONAL SERVICE

Technical reviewer for journals including Engineering Structures, Structural Concrete, Construction and Building Materials, Journal of Mechanics of Materials and Structures, Acta Geophysica, Applied Acoustics, Canadian Journal of Civil Engineering, and Nature Communications.